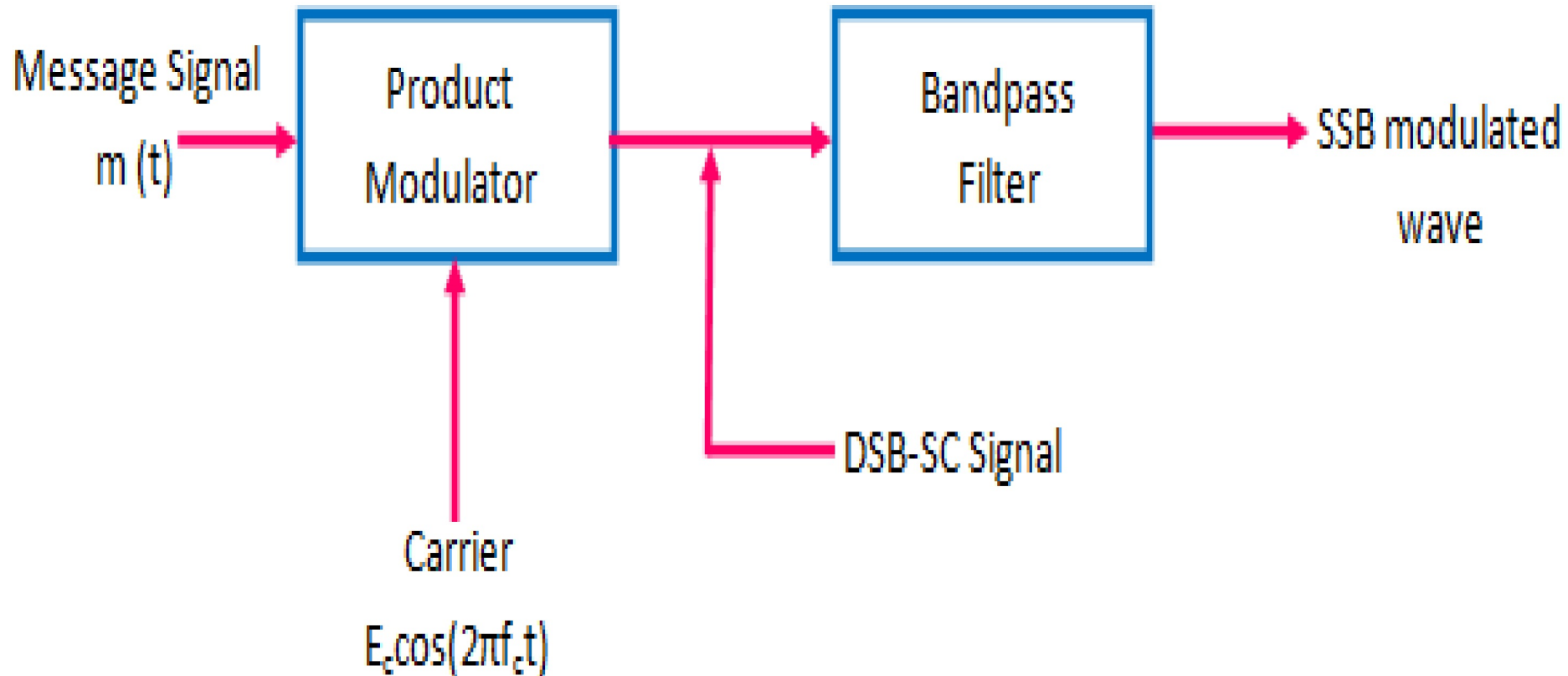




Single Side Band Systems (SSB)

Complied By:
Kartik R. Patel
EXTC Department
KJSSE, SVU

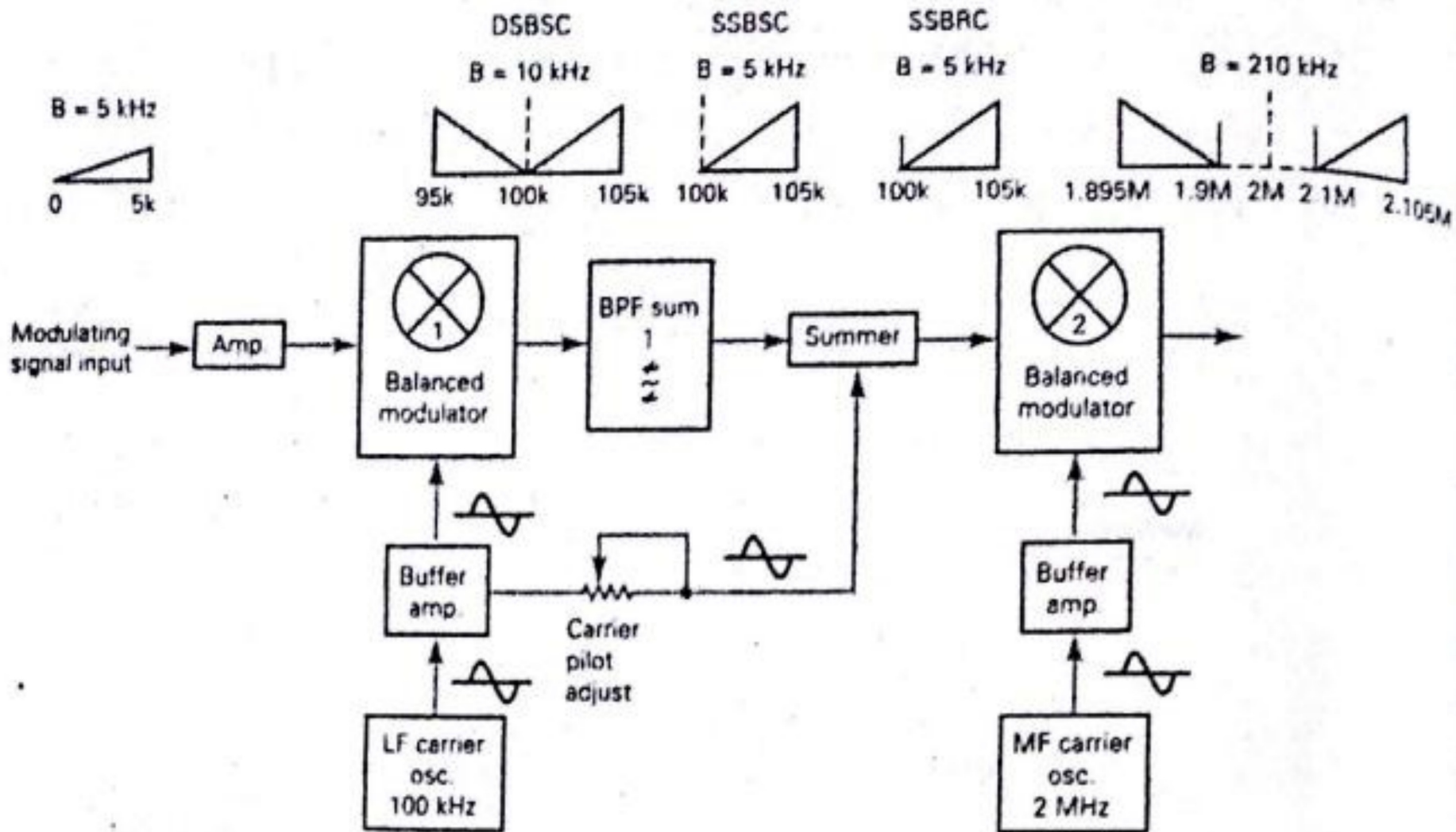
Single Sideband (SSB) Basic principle



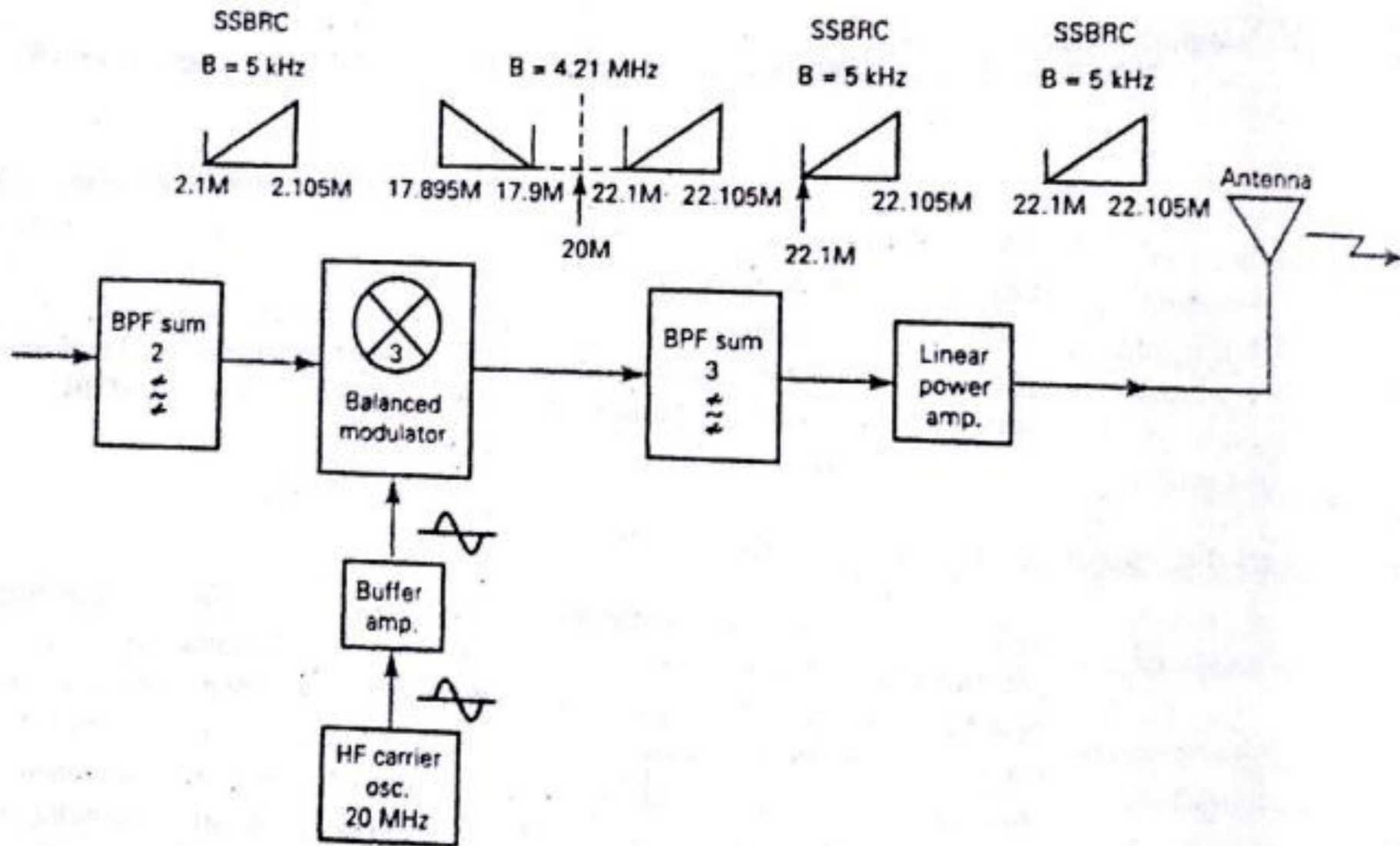
Single Sideband (SSB) generation methods

- Filter Method
- Phase shift Method
- Third Method

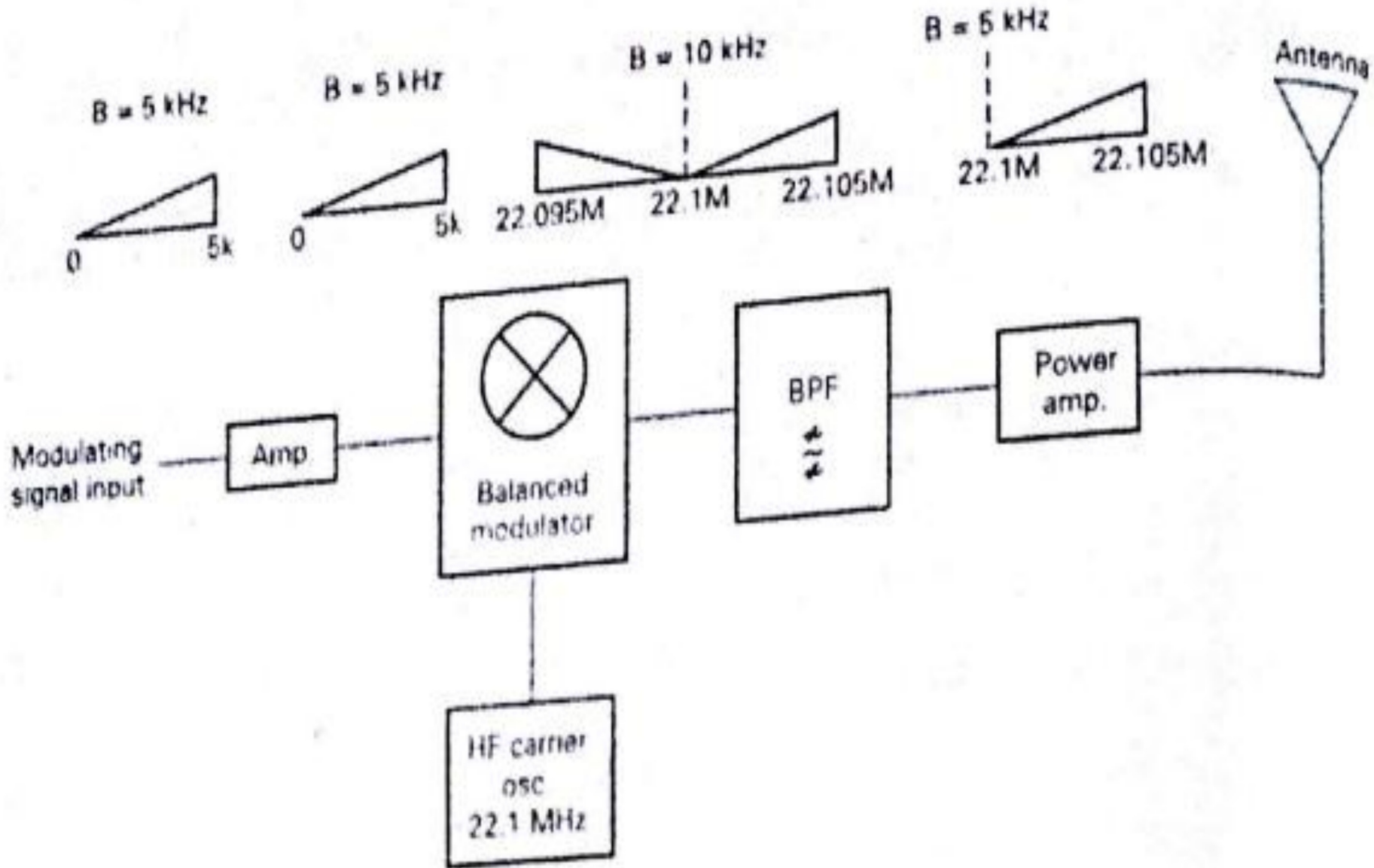
SSB Generation- Filter Method



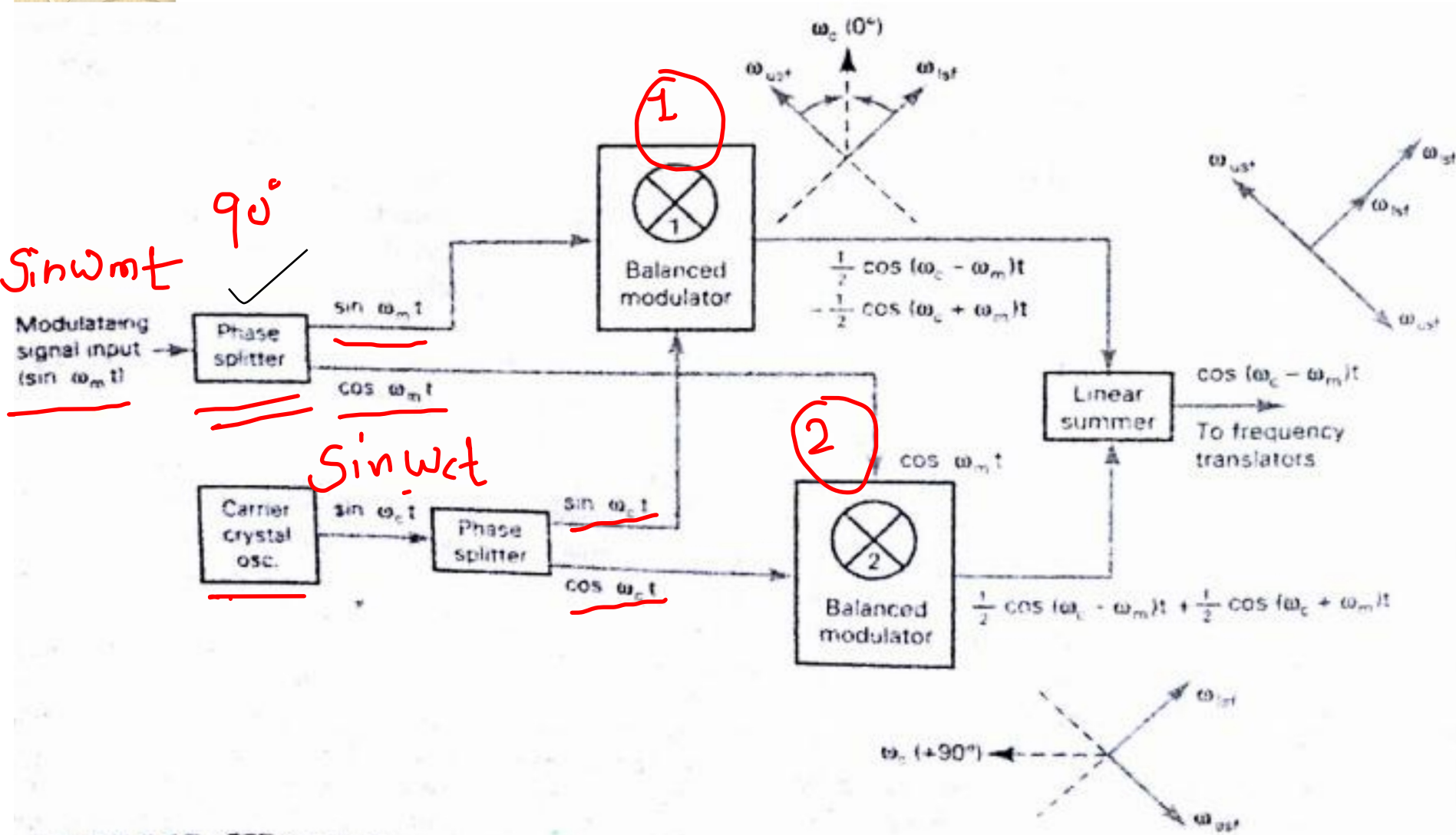
SSB Generation- Filter Method



SSB Generation- Filter Method



SSB Generation- Phase Shift Method



SSB Generation- Phase Shift Method

These outputs are expressed mathematically as

output from

balanced modulator 1 = $(\sin \omega_m t)(\sin \omega_c t)$

$\sin A \cdot \sin B$

$$= \frac{1}{2} \cos(\omega_c - \omega_m)t - \frac{1}{2} \cos(\omega_c + \omega_m)t$$

output from

balanced modulator 2 = $(\cos \omega_m t)(\cos \omega_c t)$

$\cos A \cdot \cos B$

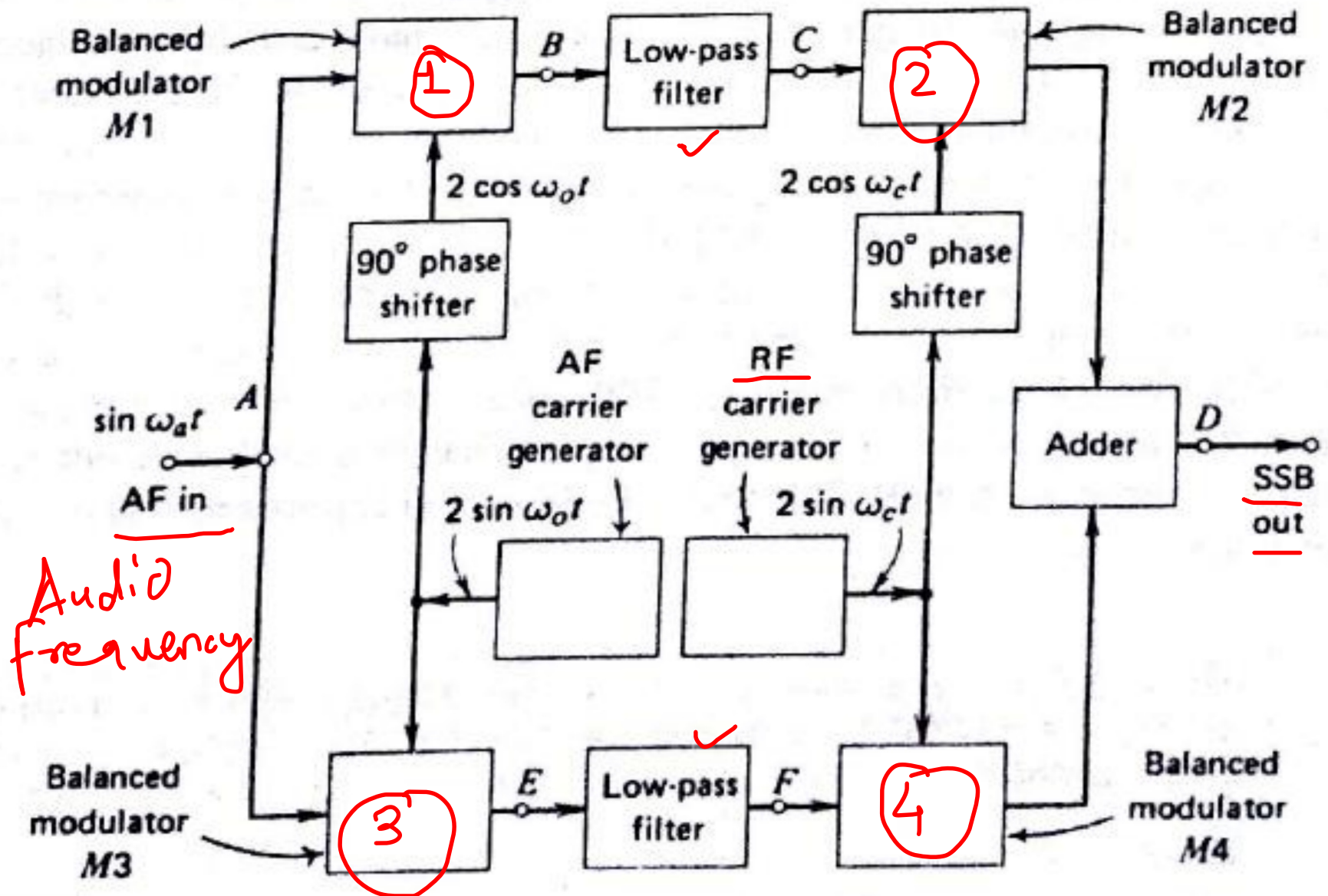
$$= \frac{1}{2} \cos(\omega_c - \omega_m)t + \frac{1}{2} \cos(\omega_c + \omega_m)t$$

and the output from the linear summer is

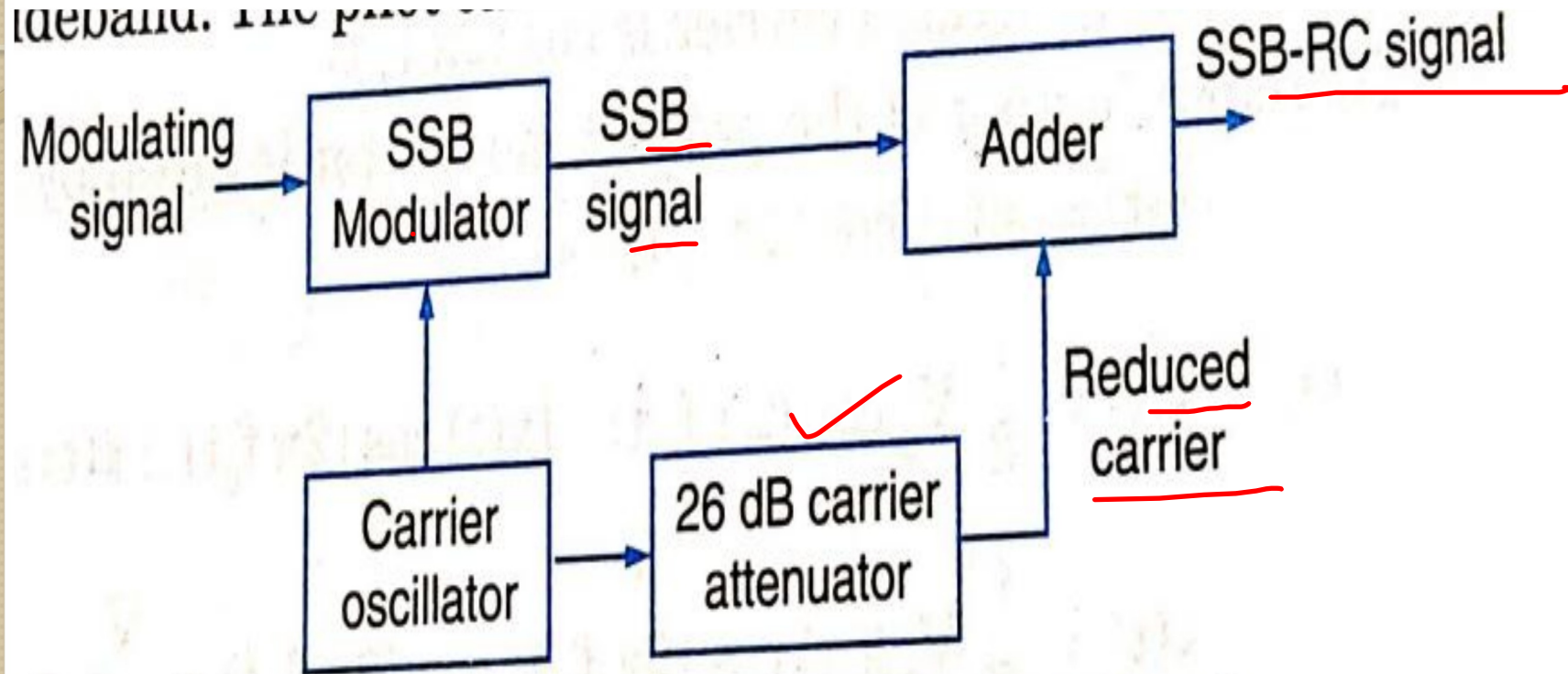
$$\begin{array}{r} \frac{1}{2} \cos(\omega_c - \omega_m)t - \frac{1}{2} \cos(\omega_c + \omega_m)t \\ + \frac{1}{2} \cos(\omega_c - \omega_m)t + \frac{1}{2} \cos(\omega_c + \omega_m)t \\ \hline \cos(\omega_c - \omega_m)t \quad \text{canceled} \end{array}$$

lower sideband
(difference signal)

SSB Generation- Third Method



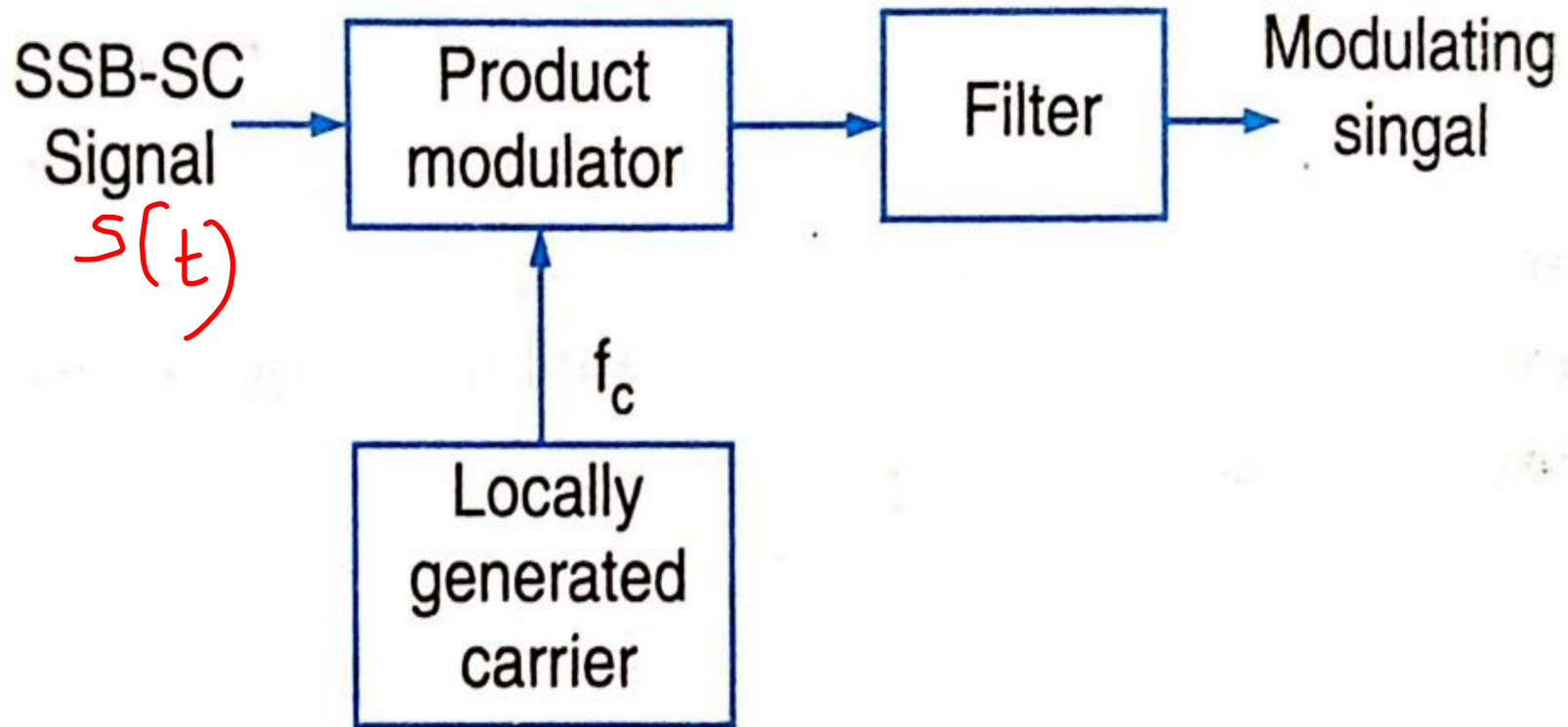
Pilot Carrier SSB System (SSB-RC)



Demodulation of SSB Waves

- **Requirements of SSB receiver**
 - High reliability
 - Excellent suppression of adjacent signals
 - High signal to noise ratio
 - Ability to demodulate SSB

Coherent SSB Demodulation



Analysis of Coherent SSB Detector

Let the SSB wave at the input be given by,

$$\rightarrow \underline{s(t)} = \frac{1}{2} V_c [\underline{x(t) \cos(2\pi f_c t)} \pm \hat{x}(t) \sin(2\pi f_c t)]$$

The locally generated carrier is $\cos(2\pi f_c t)$

Therefore, output of the product modulator is given by,

$$v(t) = \underline{s(t)} \cdot \cos(2\pi f_c t)$$

SSB Locally generated

or $v(t) = \frac{1}{2} V_c \cos(2\pi f_c t) \cdot [x(t) \cos(2\pi f_c t) \pm x(t) \sin(2\pi f_c t)]$ ✓

$$v(t) = \frac{1}{2} V_c x(t) \underbrace{\cos(2\pi f_c t) \cos(2\pi f_c t)} \pm \frac{V_c}{2} \hat{x}(t) \underbrace{\cos(2\pi f_c t) \sin(2\pi f_c t)} \quad \checkmark$$

Analysis of Coherent SSB Detector

But, $\cos A \cos B = \frac{1}{2} \{ \cos (A + B) + \cos (A - B) \}$

and $\cos A \sin B = \frac{1}{2} \{ \sin (A + B) + \sin (A - B) \}$

Note

Therefore, $v(t) = \frac{1}{4} V_c x(t) [\cos(4\pi f_c t) + \cos(0)] \pm \frac{1}{4} V_c \hat{x}(t) [\sin(4\pi f_c t) - \sin(0)]$

or $v(t) = \frac{1}{4} V_c x(t) + \frac{1}{4} V_c [x(t) \cos(4\pi f_c t) \pm \hat{x}(t) \sin(4\pi f_c t)]$

Scaled
message signal

Unwanted terms

filtered

When $v(t)$ is passed through the filter, it will allow only the first term to pass through and will reject all other unwanted terms. Thus, at the output of the filter we get the scaled message signal and the coherent SSB demodulation is achieved.

Therefore, $v_0(t) = \frac{1}{4} V_c x(t)$

... (3.119)